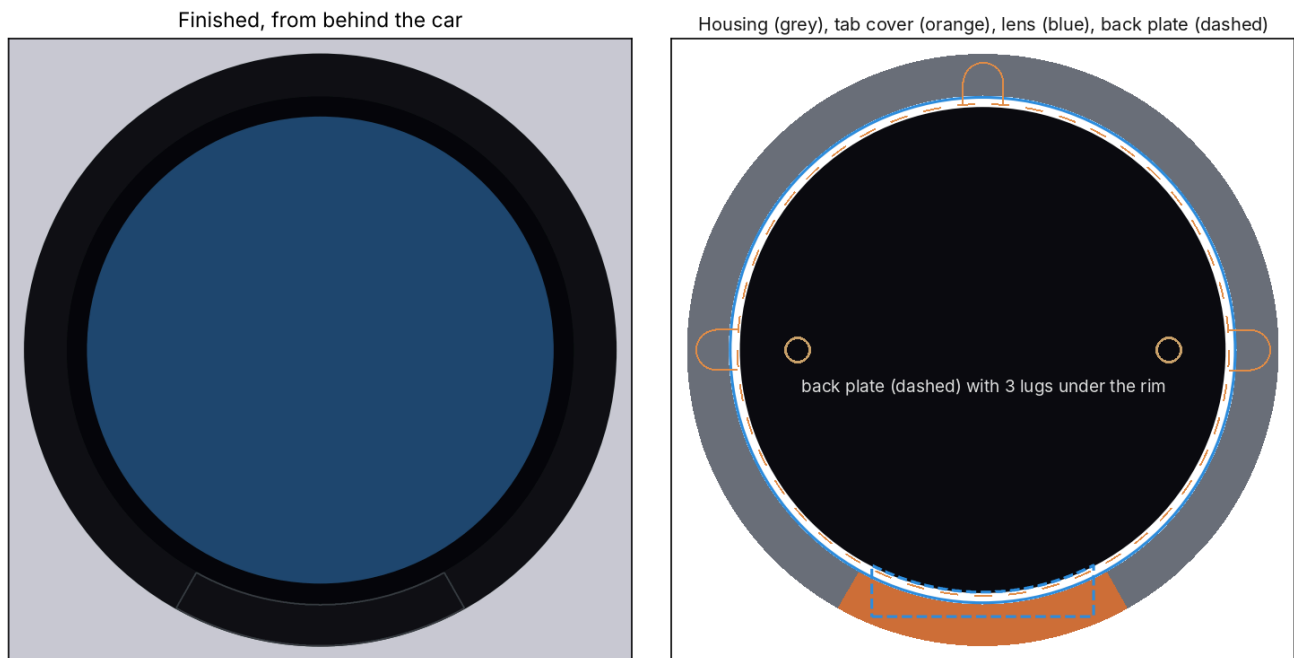


Trunk Emblem Display

A round 2.8" ESP32 touchscreen in a 3D-printed housing that replaces the 74 mm trunk roundel. It has an open back so it sits just 10.8 mm off the paint, plugs into the factory peg holes without drilling, runs from an always-on 12 V fuse with battery protection, and shows the classic roundel or your own designs, controlled from your phone.

Everything is at github.com/cakemasterjedi/haloweb

Install the firmware: cakemasterjedi.github.io/haloweb. Print files are in the **hardware** folder, and this guide and the wiring diagram are in **docs**. Green **Code** button → **Download ZIP** gets it all at once.



89.1 mm

outside diameter

10.8 mm

off the paint, including the tape

56 mm

factory peg spacing (all G30)

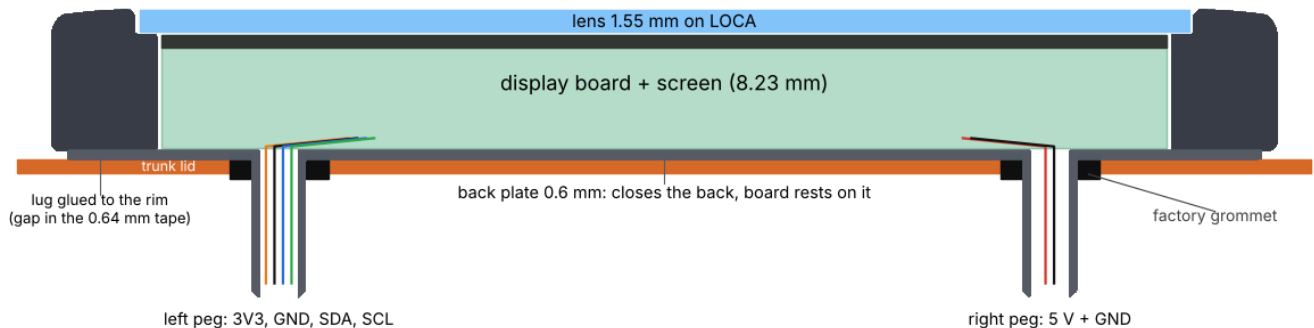
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1 Before you start

The build has two units. The **display unit** replaces the badge: a printed housing holding the Waveshare board, sealed under a clear acrylic lens. The **trunk-lid tray** sits behind the lid's inner trim and holds the 12 V → 5 V converter and the INA219 voltage sensor. A thin printed **back plate** closes in the back of the housing. It sits on the paint under the board, and three lugs on its edge reach out under the rim through gaps in the tape ring and are glued to the rim, so the plate and housing are one unit. The plate's two hollow pegs go into the factory emblem grommets, and the wires between the two units run down through them, so nothing is drilled.

Cross-section through the pegs and lugs, 9 o'clock to 3 o'clock: 10.8 mm off the paint



How the parts stack, to scale. The board and screen (8.23 mm) set most of the height. The board rests on the 0.6 mm back plate on the paint. The rim sits on 0.64 mm VHB tape at the same level, and the 1.55 mm lens sits on top.

Read these first

- **Battery:** disconnect the 12 V battery (in the trunk on the G30) before any wiring, and tap an always-on circuit only with the firmware's battery protection turned on (section 4). Fit the kill switch if the car sits for weeks.
- **Legal:** rules on lit or animated displays facing backward vary by state and country. Check yours; a dim or static image is the safe choice while driving.
- **Sealing:** the display is sealed shut with silicone. Install the firmware and check the phone control page works before step 6.9 (bonding the lens), because the USB port can't be reached afterwards.
- **Heat:** print the final parts in ASA. PLA and PETG soften on a sunny trunk lid.

Unofficial fan project, not affiliated with or endorsed by BMW. Work at your own risk.

2 Parts and tools

PART	QTY	NOTES
☐ Waveshare ESP32-S3-Touch-LCD-2.8C	1	73 mm round glass, 480×480. Comes with 12-pin and 4-pin SH1.0 cables.
☐ 3" × 1/16" clear acrylic disc	1	Enoin 3" × 0.060" (76.2 × 1.5 mm). The lens.
☐ INA219 module (GY-219)	1	25 × 22 mm. Reads the car's voltage over I2C.
☐ YRDZXG 12/24 V → 5 V 5 A 25 W buck converter	1	Sealed. Body about 46 × 32 × 18 mm, 61 mm over the mounting ears. Red/black in, yellow/black out.
☐ 28 AWG silicone wire, red + black	~1 m each	5 V and ground, right peg. Two must fit the 2.7 mm hole (≈1.0 mm OD each).
☐ 30 AWG silicone wire, 3 colors	~1 m each	3.3 V, GND, SDA, SCL (4 wires), left peg.
☐ 20 AWG automotive wire, red + black	~3 m each	12 V feed from the fuse box to the trunk lid.
☐ Fuse tap kit (Mini and ATO) + 3 A fuses	1	The G30 fuse box has both Mini and ATO (standard-size) blade fuses. Get a kit with both tap sizes and 3 A fuses in both sizes; you use the one that matches the fuse you tap.
☐ 12 V kill switch, 5 A or more (optional)	1	Inline rocker on a pigtail, or a small sealed toggle. Turns the display fully off for storage.
☐ Ring terminal (M6 / ¼")	1	Chassis ground.
☐ Solder-seal heat-shrink connectors	~10	Waterproof wire joins.
☐ LOCA UV-cure optical glue	1	Bonds the lens to the screen glass. Not B-7000 or super glue: they craze acrylic.
☐ Permatex Ultra Black (82180)	1	Sensor-safe, neutral-cure silicone for all seals.
☐ 3M VHB 5925 (0.64 mm), black, 1"	1 roll	Cut into a ring for the rim with the template in section 8. Thinner than 5952 so the rim sits level with the back plate.
☐ 3M adhesion promoter, isopropyl alcohol	—	Surface prep for the VHB.
☐ Mini silica gel pack	1	Taped to the back of the board.
☐ Conformal coating, e.g. MG Chemicals 422C (silicone) or 419D (acrylic)	1 small can or pen	Coats the boards so a drop of water or condensation can't short them. See step 6.4.
☐ Small UV-resistant zip ties (≤4.8 mm)	~8	Tray straps and harness ties.
☐ Tesa cloth tape, split loom 6–8 mm	—	Make the new wiring look factory and stop rattles.
☐ ASA filament (black) + PLA for test fit	—	Polymaker, Bambu or eSun ASA.

Tools: 3D printer (enclosed for ASA), soldering iron, wire strippers for 20–30 AWG, heat gun, multimeter, trim removal tools, painter's tape, UV flashlight (for LOCA), masking tape, USB-C cable and a computer with Arduino IDE.

3 Printing

FILE	ORIENTATION	PURPOSE
G30_emblem_display_SLIM_HOUSING.3mf	Back down (as supplied)	The housing
G30_emblem_display_SLIM_TAB_CAP.3mf	Face up (as supplied)	Curved cover over the screen's cable tab
G30_emblem_display_SLIM_BACK_PLATE.3mf	Flat, pegs up (as supplied)	0.6 mm plate that closes the back; lugs glue to the rim, hollow pegs carry the wires
trunk_electronics_tray.3mf	Flat, back down	Holds the converter and INA219 in the trunk lid

Settings (ASA)

- 0.2 mm layers (0.12 mm gives a smoother domed rim), **5 walls, 6 top / 5 bottom layers**, 40 % gyroid infill, flow about 2–5 % above your normal setting. Thick, slightly over-filled walls are what make an FDM print watertight; with the usual 2–3 walls, water can seep through the layer lines.
- Ironing on for top surfaces of the housing and tab cover so both get the same sheen.
- **No brim needed.** All parts sit flat with full bed contact and rounded outlines. Use a clean textured PEI or glue-stick bed at 100–110 °C, enclosure closed, no part cooling for the first 3 layers, and a slow first layer. No supports needed.
- Back plate: 0.2 mm layers (exactly 3 for the plate), 3 walls so the pegs print solid around the wire hole, 100 % infill. It's thin, so let it cool on the bed before removing it to keep it flat.

Test fit in PLA first

- 1 **Make a jig:** drill two 5 mm holes 56 mm apart through a flat scrap board. It stands in for the trunk lid during the test fit and assembly. Push the back plate's pegs into it, and push a pair of 28 AWG wires through one peg to check they fit.
- 2 Set the housing over the plate on the jig with the three lugs under the rim, and drop the display in. It should rest on the plate with the glass about 0.3 mm below the lens ledge and the cable tab in the bottom pocket.
- 3 Set the acrylic disc in the rim pocket. It should sit flush with the rim. Disc thickness varies a little; if it sits proud or low, note how much.
- 4 Place the tab cover. It should sit flush, with the ribbon tucked into the pocket under it.
- 5 Hold the housing against the trunk lid. If you see a gap at the edges from the lid's curve, note roughly how big it is; the tape and plate thicknesses can be changed to match. The 0.64 mm tape fills less of a gap than thicker tape, so check this before gluing.

Adjusting the fit: every dimension is a parameter in G30_emblem_display_housing.scad; change it and export a new STL.

Factory-gloss finish (optional): wet-sand 400 → 800 → 1500, adhesion promoter, filler primer, gloss black, 2K clear, with the housing and tab cover painted together. Keep paint out of the lens and tab pockets.

4 Firmware

No coding or software needed: it installs from a web page in about a minute. Do this on the bench, before the board goes into the housing.

Install it (once)

- 1 On a **computer** (Windows, Mac, Linux or Chromebook), open **Chrome** or **Edge**. Safari, Firefox and phones can't do this.
- 2 Go to **cakemasterjedi.github.io/haloweb**.
- 3 Plug the display into the computer with a **USB-C data cable**. Some cheap cables only charge; if nothing happens, try another cable.
- 4 Click **Install**. A box pops up with a list of ports: pick the one that appeared when you plugged in (it usually has *USB* or *CH343* in the name), click **Connect**, then **Install**.
- 5 Wait for *Installation complete* (about a minute). The display restarts with the emblem on it.

No port in the list? Install the **CH343** USB driver from wch-ic.com, or unplug, hold **BOOT**, plug back in, let go and try again.

Control it from your phone

- 1 Power the display. The first time, it makes its own Wi-Fi network (**EMBLEM-** and four characters) with a temporary password, and shows both on its screen.
- 2 Join it on your phone with that password. The control page opens by itself; if not, open the browser and go to **<http://192.168.4.1>**. On Android, tap **Stay connected** when it says the network has no internet.
- 3 It asks you to choose your own Wi-Fi password (and a name if you like), then restarts its Wi-Fi. Rejoin with your new password (write it down) and pick a design and brightness.

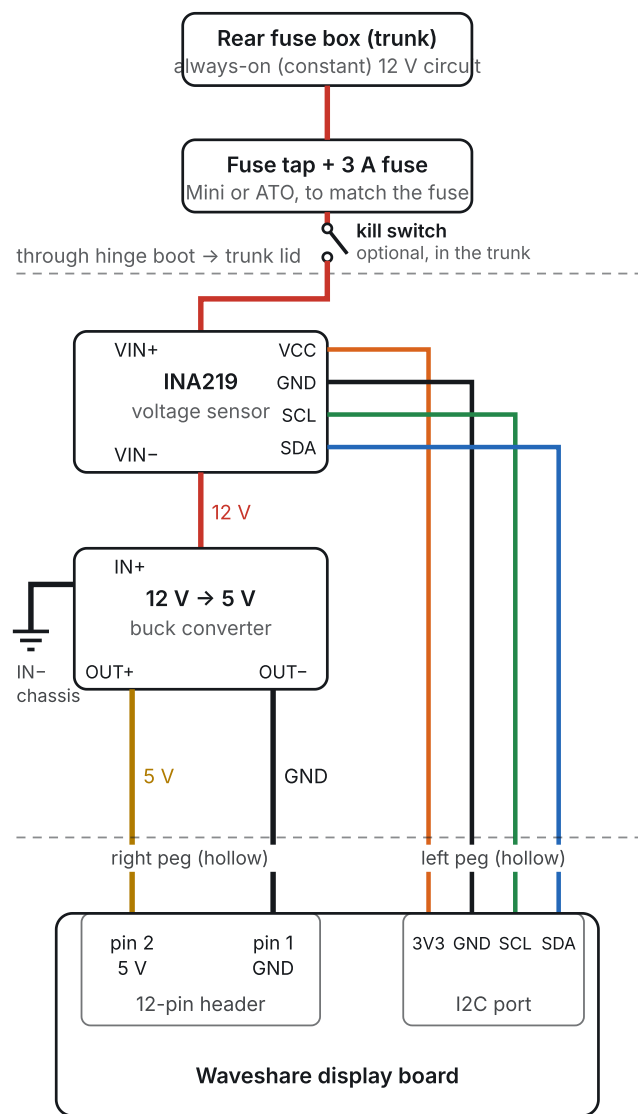
Set up the battery protection

On the **Power** tab, under **Battery protection**:

SETTING	WHAT TO CHOOSE
Voltage source	Car battery (INA219) . The live voltage appears once the INA219 is wired (section 5).
Battery protection	On , cutoff about 12.2 V . If the voltage stays below it for 30 seconds the display turns off. It checks every 10 minutes and comes back on at 0.3 V above the cutoff (engine running or charger on).
Calibrate	Type in what a multimeter reads across the battery (section 9).
Power mode → Turn off after	Leave it on Never unless you fitted a kill switch: once off, it stays off until the BOOT button on the board is pressed (sealed inside) or the kill switch cycles its power. Car show: Until the battery is low .

Updating later: download the update file from **cakemasterjedi.github.io/haloweb** on your phone, join the emblem's Wi-Fi, and install it under **Settings → Firmware**.

5 Wiring overview



12 V passes through the INA219 before the converter. Power goes up the right peg, sensor wires up the left.

FROM	TO
● Fuse tap (3 A) → kill switch, if fitted	INA219 VIN+
● INA219 VIN-	Converter IN+ (red)
● Converter IN- (black)	Chassis ground stud
● Converter OUT+ (yellow)	12-pin header pin 2 (5 V)
● Converter OUT- (black)	12-pin header pin 1 (GND)
● INA219 VCC	I2C port 3V3
● INA219 GND	I2C port GND
● INA219 SCL	I2C port SCL (GPIO7)
● INA219 SDA	I2C port SDA (GPIO15)

Never

power the INA219 from the converter's 5 V (it would put 5 V on the ESP32's data pins), and never connect 12 V to the board's BAT connector.

Left and right are as seen from behind the car. Follow the pin labels printed on the board, not the wire colors of the supplied plug cables.

6 Build the display unit

- 1 **Make the cables.** Solder 28 AWG red and black to the 12-pin plug cable's pin 2 (5 V) and pin 1 (GND). Solder four 30 AWG wires to the 4-pin plug cable's 3V3, GND, SDA and SCL. Leave about 1 m on each, insulate every joint with heat shrink, and label the far ends.
- 2 **Thread the pegs.** Push the red and black power wires down through the right peg and the four sensor wires through the left one (as seen from behind the car), from the flat side of the plate. Leave about 15 cm on the board side.
- 3 **Plug in.** Plug the wires into the 12-pin header and the I2C port. Route them flat across the back of the board so they don't sit on the brass studs, and keep them in the gaps between parts so the board still sits flat on the plate.
- 4 **Conformal-coat the board.** With both plugs pushed fully home, mask the USB-C port and the BOOT and RESET buttons with tape and keep the coating off the front glass. Then brush or spray a thin coat over the back of the board, the two plugged-in connectors and the first centimetre of the wires. Let it dry as the can says (usually 1–24 h) and peel the masking.
- 5 **Join the plate to the housing.** Put the plate on the jig from step 3.1, pegs in the holes. Put a small dab of black silicone on top of each of the three lugs, then lower the housing over the plate so the lugs sit under the rim at 3, 9 and 12 o'clock and the cable tab pocket is at 6 o'clock. Press it down so the rim sits on the lugs, and let the silicone set. Tape a mini silica gel pack to the back of the board.
- 6 **Drop the display in** from the front, with the cable tab in the bottom pocket. It rests on the plate with the glass about 0.3 mm below the lens ledge, exactly as it will on the car.
- 7 **Seal the glass edge.** Run a thin bead of black silicone into the gap between the glass and the rim. Seat the tab cover on its pocket with a little silicone on its seams, tape it down lightly, and let it cure for 24 h.
- 8 **Final firmware check.** Power the board through the new wires (5 V supply or the converter) and confirm the screen, the voltage reading and the phone control page all work.
- 9 **Bond the lens.** Clean the glass and the disc with isopropyl alcohol. Put a coin-sized pool of LOCA in the middle of the glass, lower the disc into the rim pocket from one edge so the glue spreads without bubbles, and press gently until it reaches the edges. Wipe off the excess, then cure with UV light for the time given on the bottle.
- 10 **Seal the lens edge** with a fine bead of silicone in the 0.3 mm gap and let it cure. With the back open, the lens and the glass-edge silicone are what hold the display in the housing, so handle it by the rim until it's mounted.
- 11 **Seal the pegs.** Fill each peg around its wires with silicone from the open end so water can't run down them.
- 12 **Water test before it goes on the car** (after 24 h of curing). Fit a test ring of tape on the rim, stick the unit face up on a smooth tile and fill a bowl around it to 2 cm below the top of the rim for 30 minutes, then spray it with a garden hose for 2 minutes from all sides. Power it up and check: no fog under the lens, silica gel still dry. Peel the test tape and fit fresh tape for the car.

How it's sealed

Every way in is closed: the lens is bonded to the glass with LOCA and its edge is sealed with silicone; the glass edge and the tab cover seams are silicone; on the car, the tape ring plus silicone in the lug gaps seals the rim to the paint; the pegs are filled with silicone; and the board is conformal-coated with a silica gel pack inside for any trapped moisture.

Washing

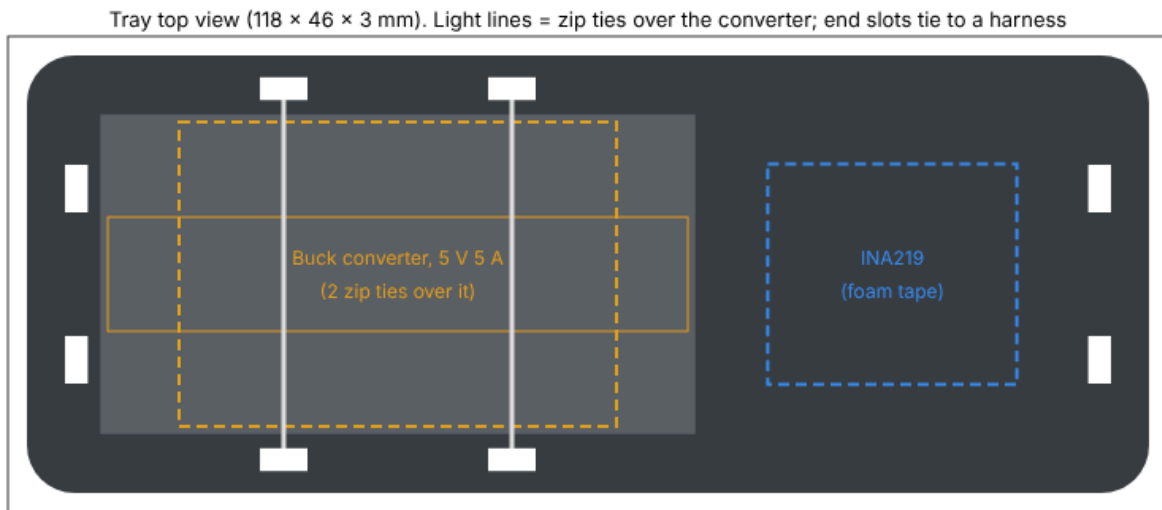
Hand washing, rain and touchless or brush car washes are fine. With a pressure washer, use a 25° or 40° (green or white) tip, stay at least 30 cm (1 ft) away and keep the jet moving; don't hold it on the rim's edge, and never use a 0° (red) tip. Wait 24 h after fitting before the first wash.

About ratings: this build follows the sealing practice used for IP67 (immersion) and pressure-washer-proof parts, and the water test above checks your unit. It isn't certified: an official IP67 or IPX9K (high-pressure, hot-water jet) rating needs testing by a lab.

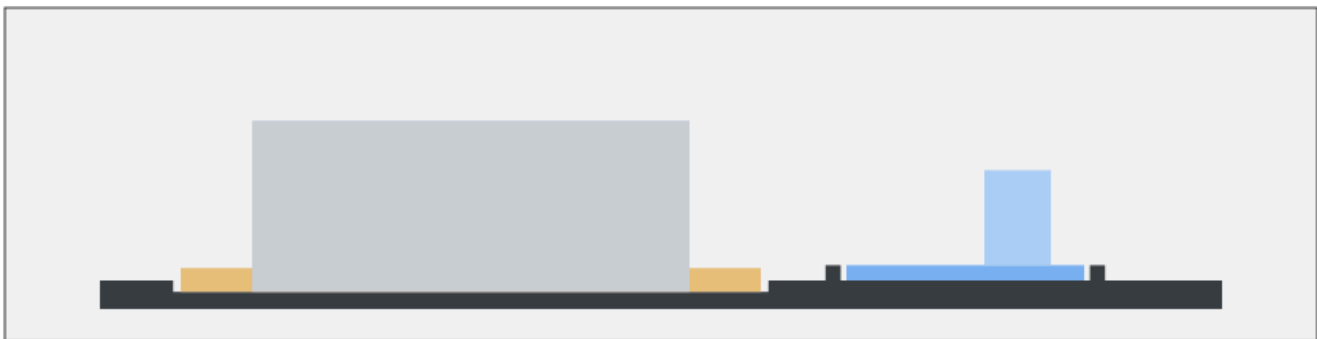
Don't use B-7000 on the lens

Its solvents can craze acrylic and fog the inside of the sealed housing. It's fine for the tab cover seam, but let it air out for a day before mounting.

7 Build the trunk-lid tray



Side view with parts in place



Converter strapped down with two zip ties; INA219 in its pocket on foam tape; end slots tie the tray to the trunk-lid harness.

- 1 Sit the converter's base and mounting ears in the pocket and strap it down with two zip ties through the slots beside it.
- 2 Solder wires to the INA219 with the pins or wires facing up, then stick it in its pocket with foam tape.
- 3 Wire the 12 V path: incoming 12 V to VIN+, VIN- to the converter's red input lead, converter's black input lead to a ring terminal for ground. Use solder-seal connectors and heat shrink everywhere.
- 4 Join the converter's yellow and black output leads to the display's power pair, and the INA219's VCC, GND, SDA and SCL to the display's sensor wires. Each wire goes to its match: the INA219's GND to the sensor GND wire.

8 Install on the car

Remove the badge and prepare the paint

- 1 Open the trunk and remove the trunk lid's inner trim panel (it clips off) to reach the back of the badge and the lid harness.
- 2 Release the badge's pegs from inside if you can reach them, or pry the badge off from outside with a plastic tool. Leave the two grommets in the holes.
- 3 Check the grommets sit flush with the paint. If their heads stand proud, the plate will rock; trim them flush.
- 4 Clean the area with isopropyl alcohol, then wipe adhesion promoter where the tape ring will sit.

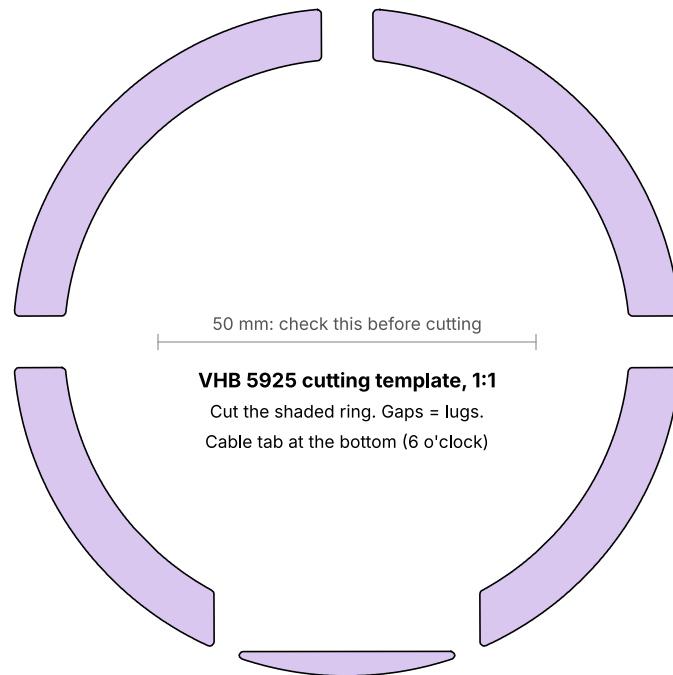
Run the 12 V feed

- 1 Find an **always-on** fuse in the trunk fuse box (right side, behind the trim cover). Lock the car and wait about 20 minutes for it to go to sleep. Clip a test light or the multimeter's black lead to a bare metal bolt, then touch the two little metal test points on top of each fuse. A fuse that still lights up (or reads about 12 V) on one side is always on.
- 2 Good ones to test first are the **12 V socket** fuses (shown as "cigarette lighter / 12 V socket" on the fuse chart) and slots for features your car doesn't have. Never tap anything labelled DME, IVM, airbag, seatbelt, fuel pump, steering column or DSC.
- 3 Pull the fuse you chose and check its size: **Mini** (small) or **ATO** (standard, larger). Use the matching fuse tap and a 3 A fuse of the same size. Disconnect the battery. Fit the fuse tap with the original fuse in the lower slot and a 3 A fuse in the upper slot, turned so the new lead is on the load side.
- 4 **Optional kill switch:** splice it into the red lead right after the fuse tap and leave it where you can reach it, behind the trunk side trim flap near the fuse box. An inline rocker switch on a pigtail needs no mounting.
- 5 Ring-terminal the ground to a factory ground stud on bare metal.
- 6 Run the red 20 AWG lead along the factory harness to the trunk hinge and through the rubber boot into the lid, inside split loom, wrapped in Tesa tape. Keep it clear of the hinge, latch and any sharp edges.

Mount the tray and the display

- 1 Zip-tie the tray to the harness inside the lid (or VHB it to a clean inner panel), somewhere you can reach later. Connect the 12 V feed to the INA219's VIN+.
- 2 **Cut the tape.** Print the template below at 100 % scale (check the 50 mm line), lay the 0.64 mm VHB over it and cut the pieces. The gaps are for the three lugs.
- 3 Stick the pieces on the housing's rim around the lugs. Fill each lug gap and the two cable-tab corners, where there's no room for tape, with a thin bead of black silicone so the ring is sealed all round.
- 4 Feed the wires through the grommets into the lid. Peel the liners, push the pegs into the grommets (cable tab at 6 o'clock), and press the housing firmly all round for 30 seconds.

- 5 From inside, join the wires to the tray as in step 7.4.
- 6 Secure the loose wiring inside the lid with tape and zip ties, then refit the trim panel. Leave the car for 24 h before washing so the tape reaches full strength.



VHB cutting template, 1:1. Print at 100 % (no "fit to page") and check that the line measures 50 mm. Also in the package as VHB_tape_template.svg.

9 Test and calibrate

- 1 Reconnect the battery and wake the car. The display should boot within a few seconds.
- 2 With the engine off, measure the battery with a multimeter. Join the emblem's Wi-Fi on your phone, open the **Power** tab, type the reading into **Calibrate** and tap Calibrate. The page and the screen now match the meter.
- 3 Start the engine: the reading should rise to about 13.6–14.6 V and show "Charging".
- 4 Lock the car and leave it. The display should stay on. If you want to test the battery protection, raise the cutoff above the battery voltage for a moment: after 30 seconds the display turns off, and it comes back within 10 minutes of starting the engine. Set the cutoff back afterwards.
- 5 If you fitted the kill switch, check it turns the display fully off.

READING	WHAT IT MEANS
13.2 V and up	Engine running, alternator charging
12.4–12.8 V	Healthy battery at rest
12.0–12.4 V	Partly discharged
Under 12.0 V	Low; under 11.8 V for a minute triggers the optional backlight-off action

10 Troubleshooting

PROBLEM	CHECK
Display doesn't turn on	Fuse tap direction and 3 A fuse; 5 V at the converter output; pin 1 / pin 2 on the 12-pin header the right way round.
Voltage shows "No sensor"	INA219 VCC on 3.3 V (not 5 V); SDA/SCL not swapped; plugged into the I2C port, not the UART port.
Fog inside the lens	A seal is leaking or the housing was closed on a humid day. Replace the silica gel and reseal the glass edge. Assemble on a dry day.
Screen goes dark on a hot day, comes back later	That's the heat guard: it dims the screen from about 70 °C on the chip and turns it off above 85 °C until it cools to 75 °C. The Power tab shows the chip temperature and the hottest it has been. The ESP32-S3R8 on this board is rated to 65 °C air temperature and the LCD to 70 °C (80 °C when off).
Display goes dark when the car is off	The tapped fuse is switched, not always-on: pick another (section 8). Or the battery is under the cutoff; charge it. Or Normal mode auto-off is on: turn it off.
Battery runs down while parked	Raise the battery protection cutoff, or use the kill switch when the car will sit. Asleep, the display still draws a little.

Can't find the emblem's Wi-Fi	The display may be off (battery protection or auto-off). Turn it off and on with the kill switch or fuse. Look for the name you chose at setup (or EMBLEM-xxxx on a new display).
Web installer finds no port	Use Chrome or Edge on a computer and a USB-C data cable. Hold BOOT while plugging in, then try again.
Tab cover sits high	The ribbon needs more room. Increase cap_recess in the .scad file and reprint the cover.

A Center-hole option

If the factory grommets are damaged or you'd rather have no pegs, the same housing can be mounted on tape alone with the wires going through one 16 mm hole drilled behind the screen, the way the Vusi Studios Dot is installed. Use G30_emblem_display_SLIM_BACK_PLATE_CENTER_HOLE.3mf instead of the peg plate: the same plate with a 12 mm center hole and no pegs. Everything else is built the same way.

- **Drilling:** 16 mm (5/8"), centered between the two peg holes (28 mm from each). Check behind the panel first for braces, wiring, the camera and the latch. Deburr, vacuum the chips and paint the bare metal edge.
- **Cables:** both go through a 5/8" rubber grommet, sealed with silicone. Any wire size works.
- **Mounting:** VHB ring only. Remove the peg grommets and seal the two peg holes flush.